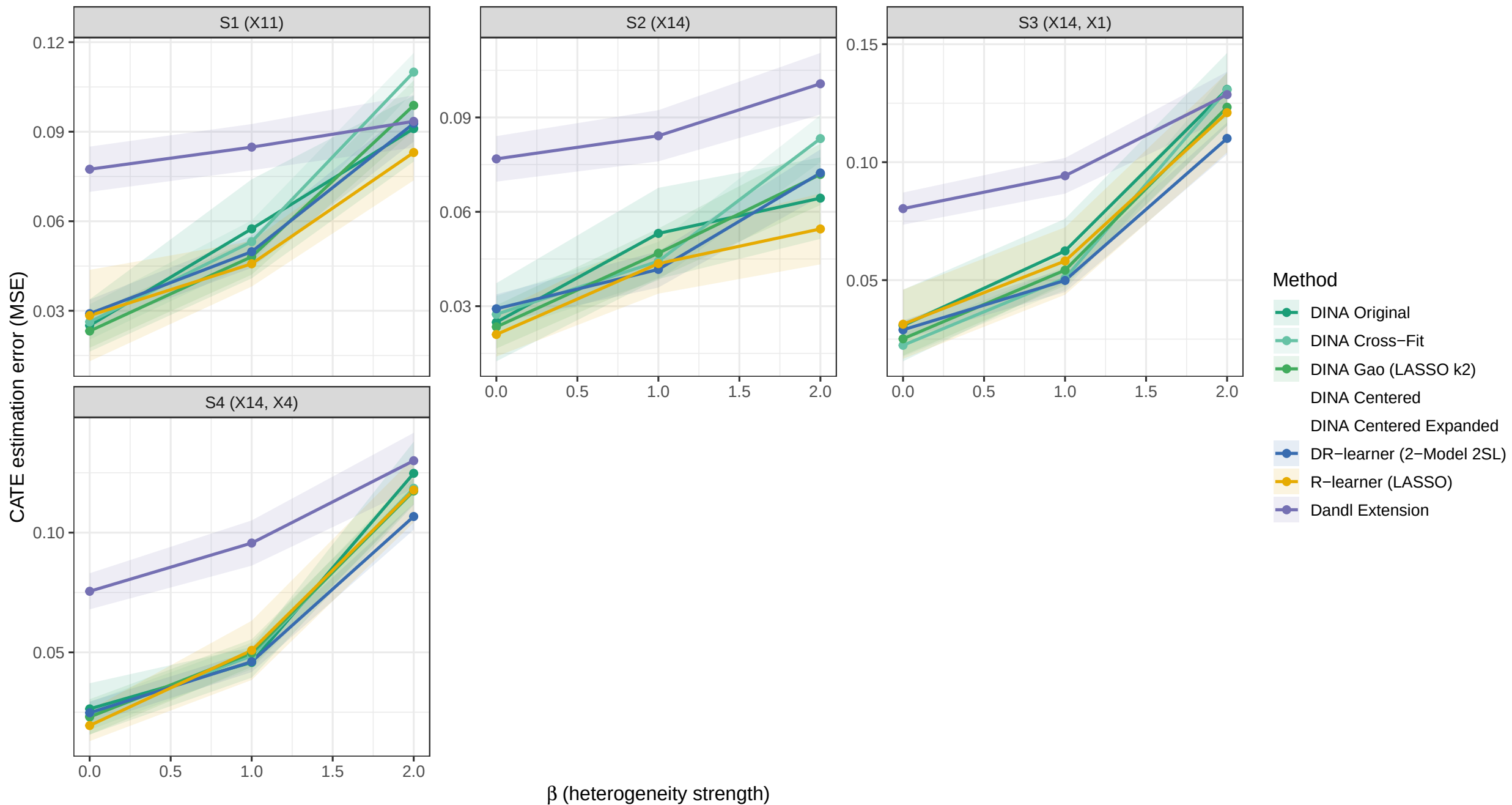


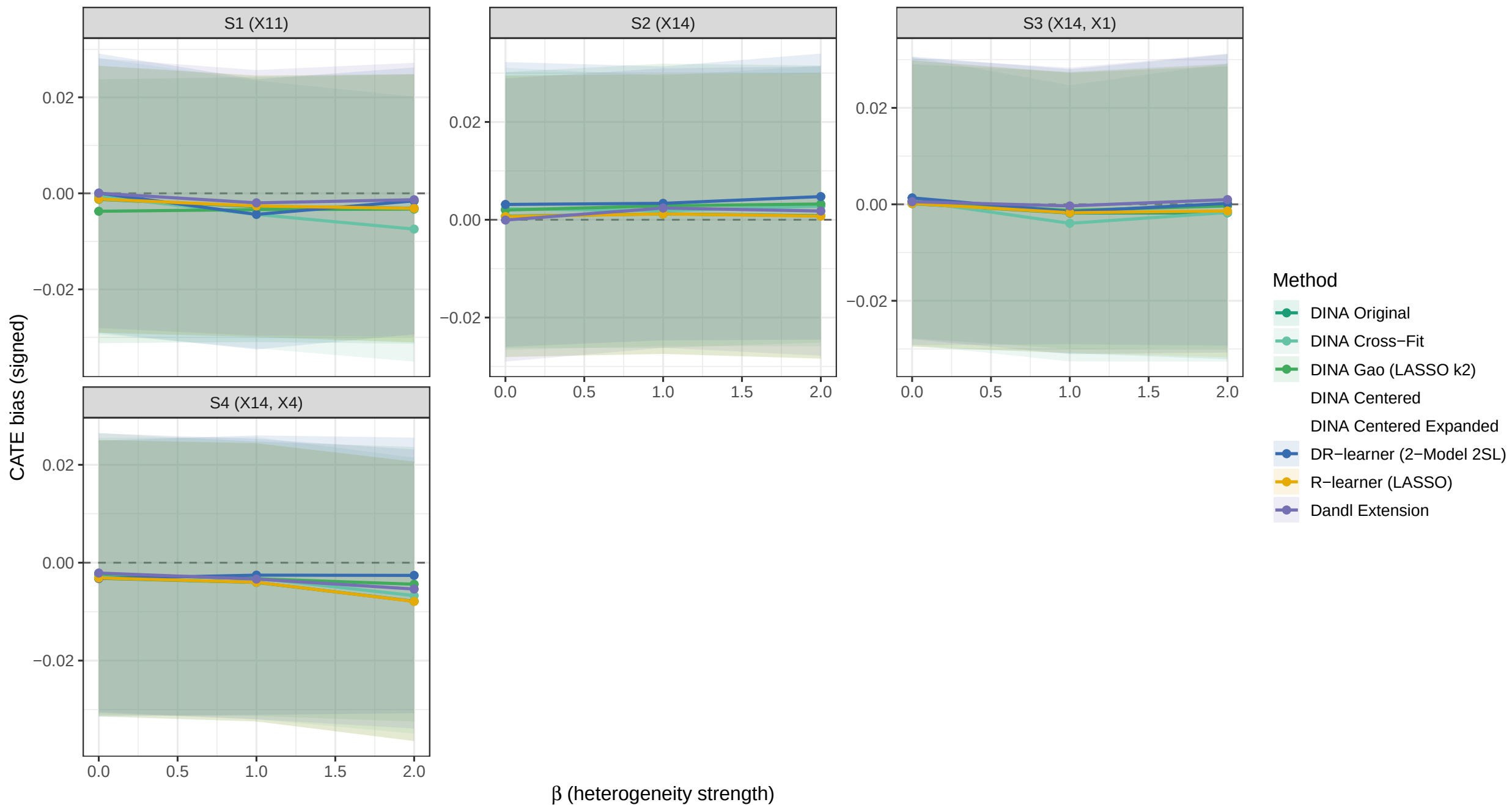
CATE estimation error (MSE)

Lower is better. Ribbons = 95% Monte-Carlo CI. $\beta=0$ is the null (constant effect).



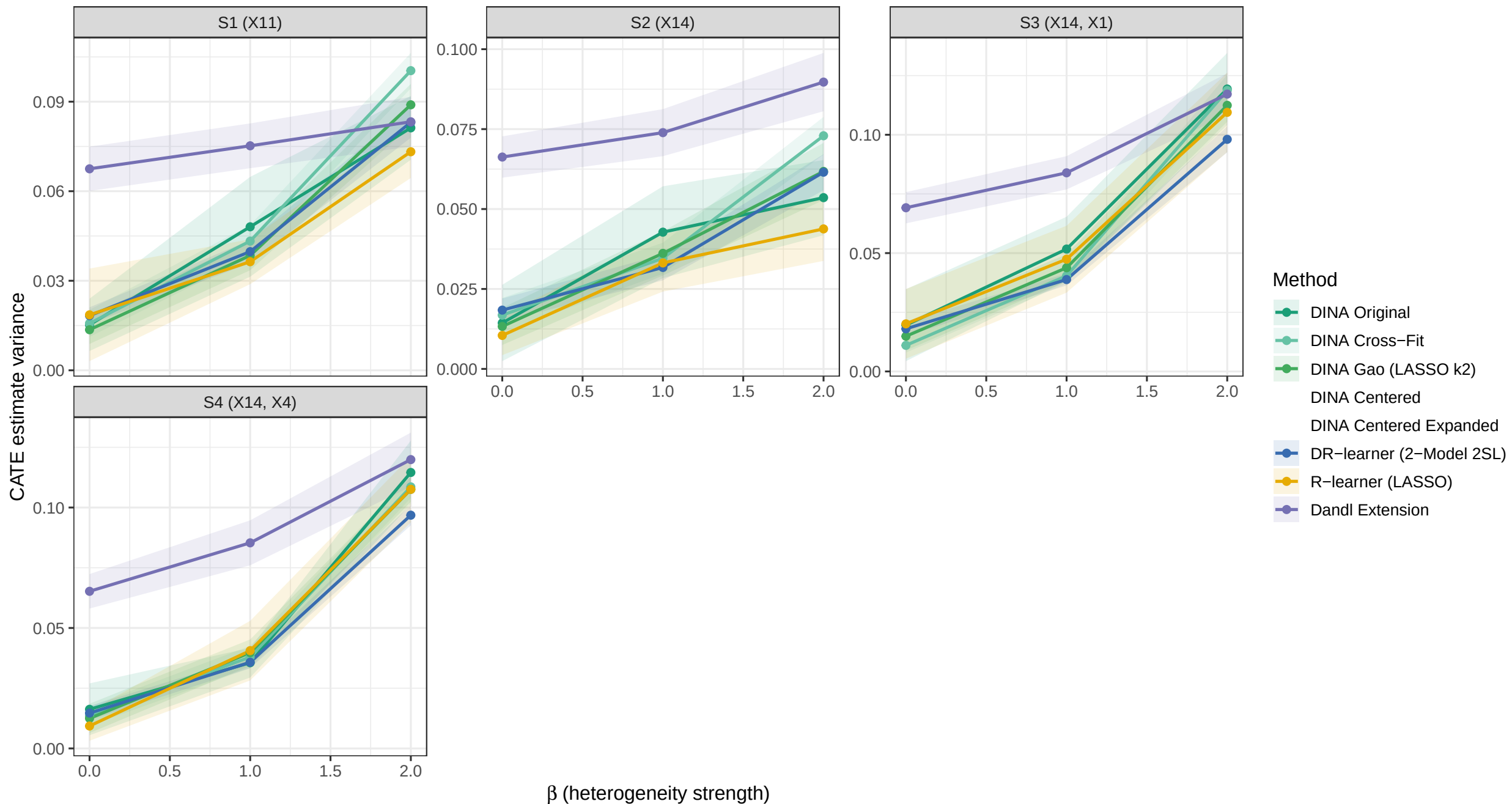
CATE bias (signed)

Closer to 0 is better. Dashed line = 0.



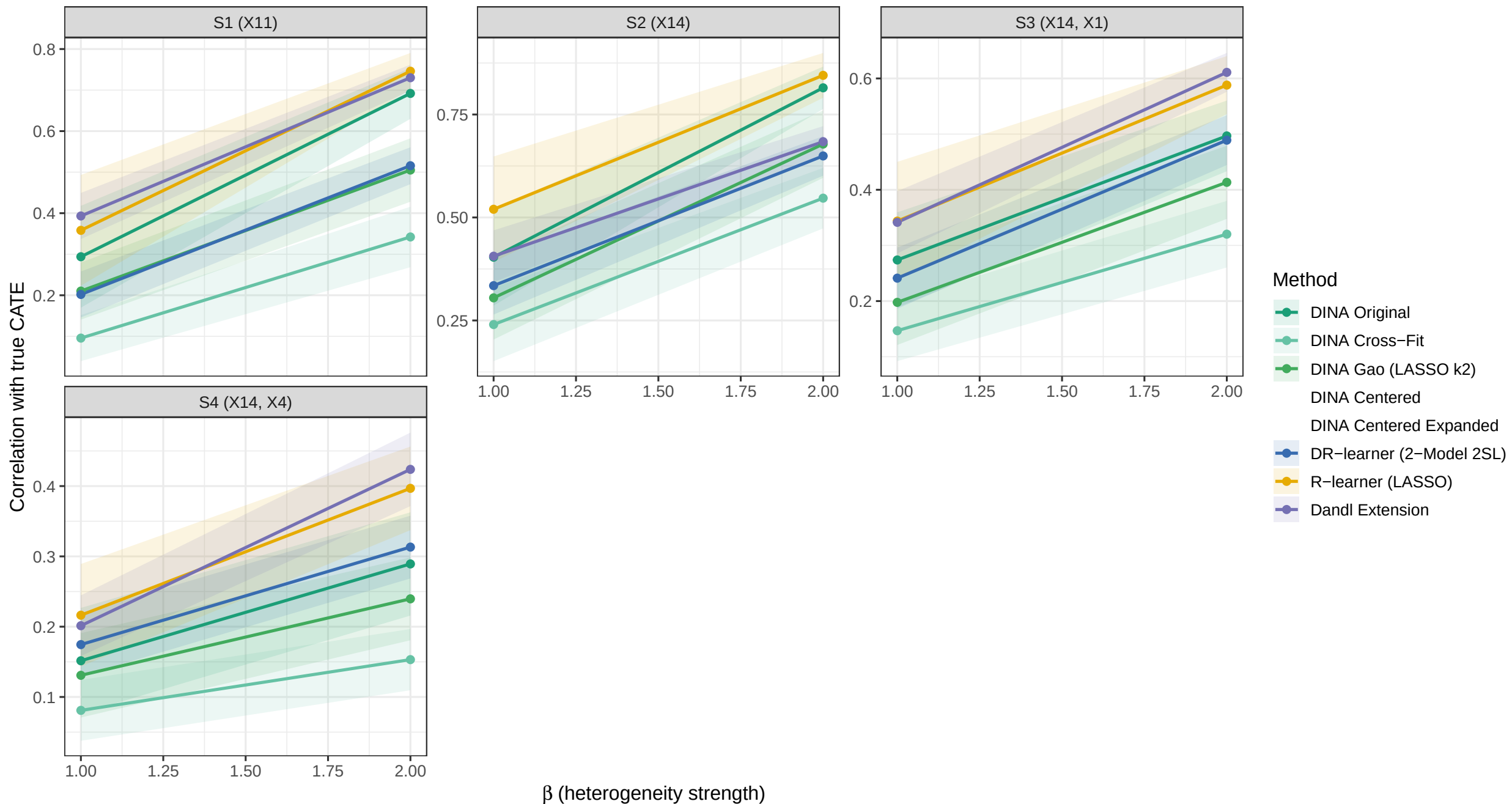
CATE estimate variance

Lower = more stable across the sample. Ribbons = 95% MC CI.



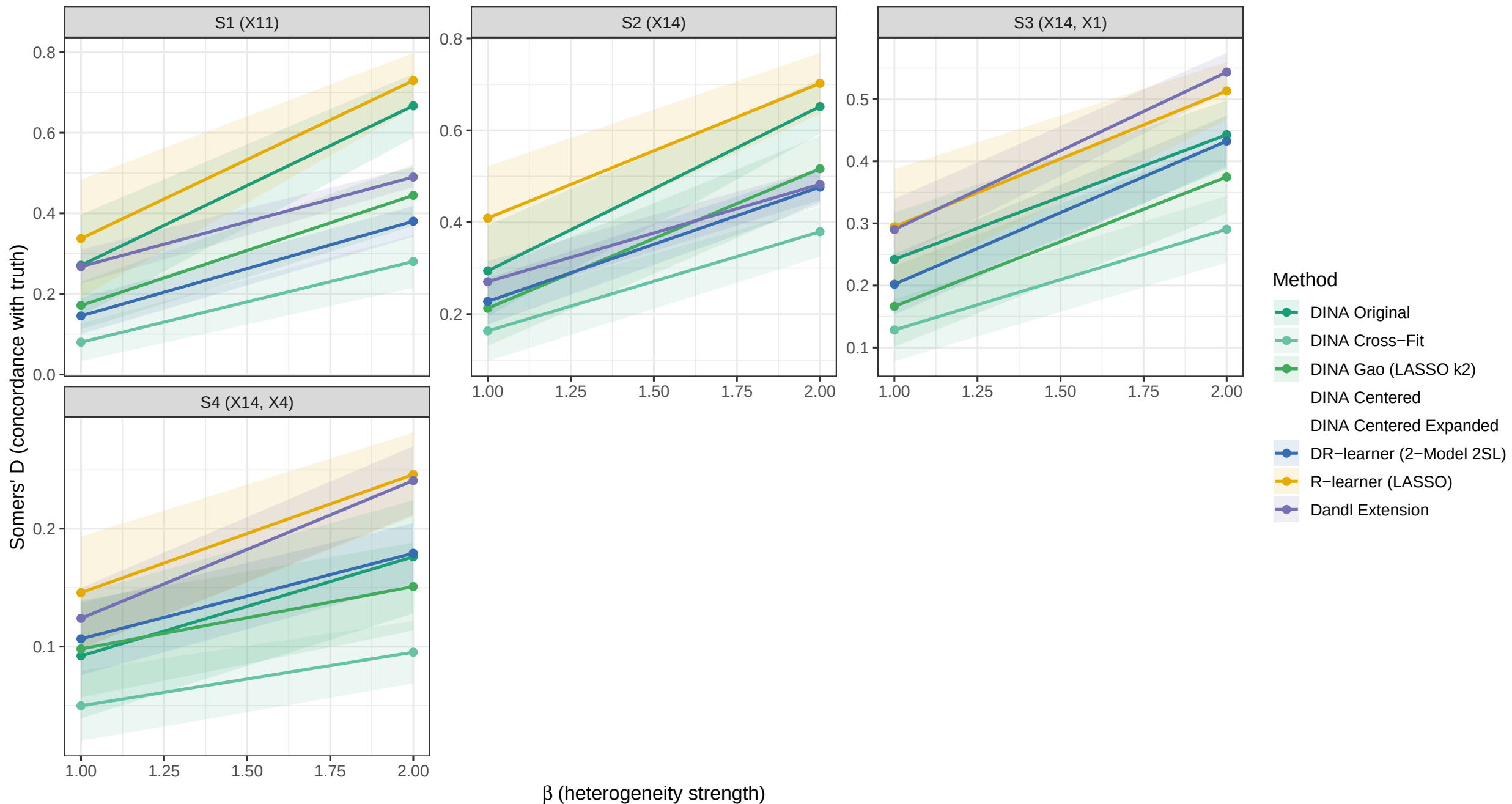
Correlation with true CATE

Higher is better (ranking quality). Undefined at beta=0, so shown for beta>0.

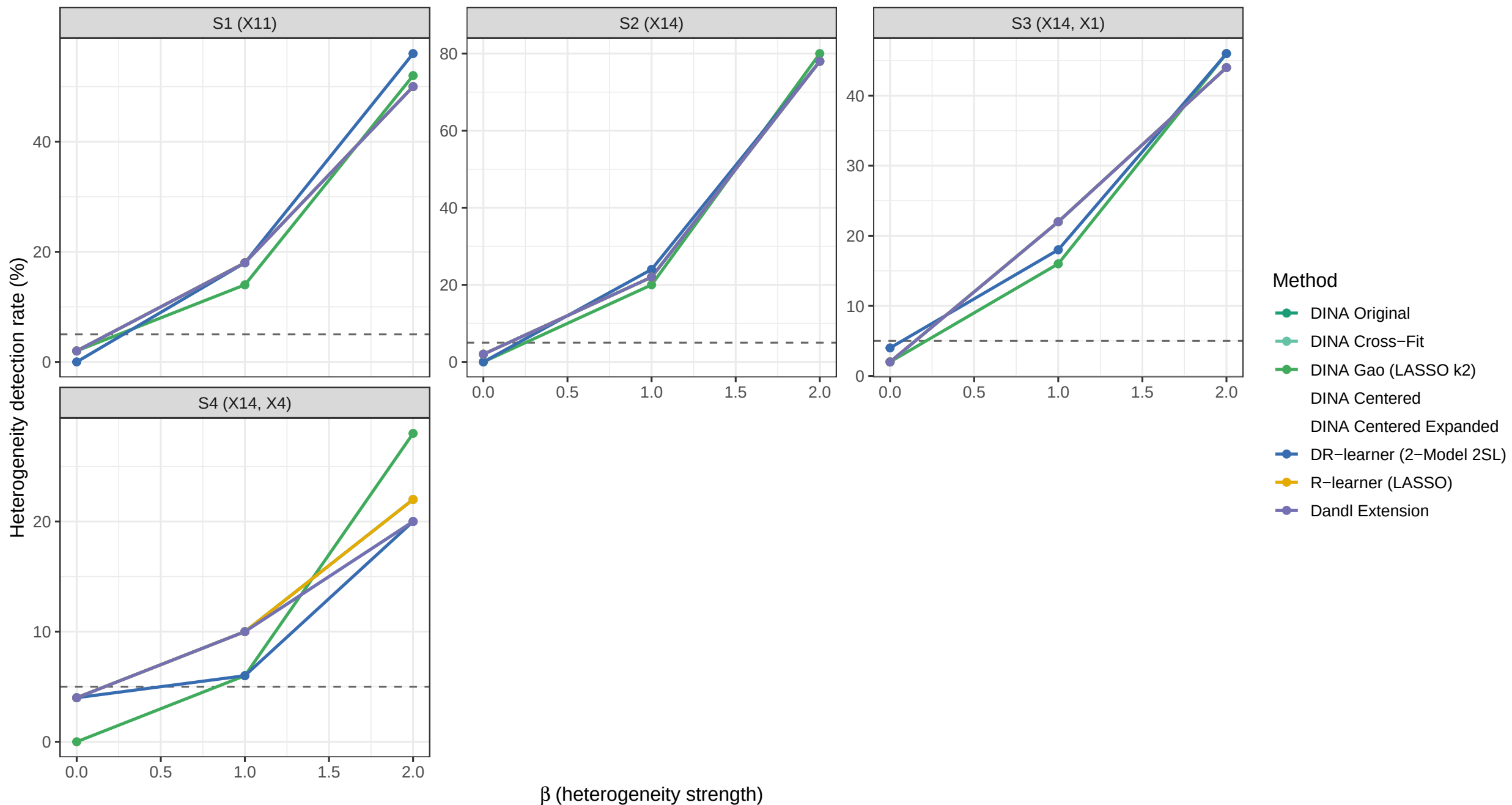


Somers' D (concordance with truth)

Higher is better. Shown for $\beta > 0$.

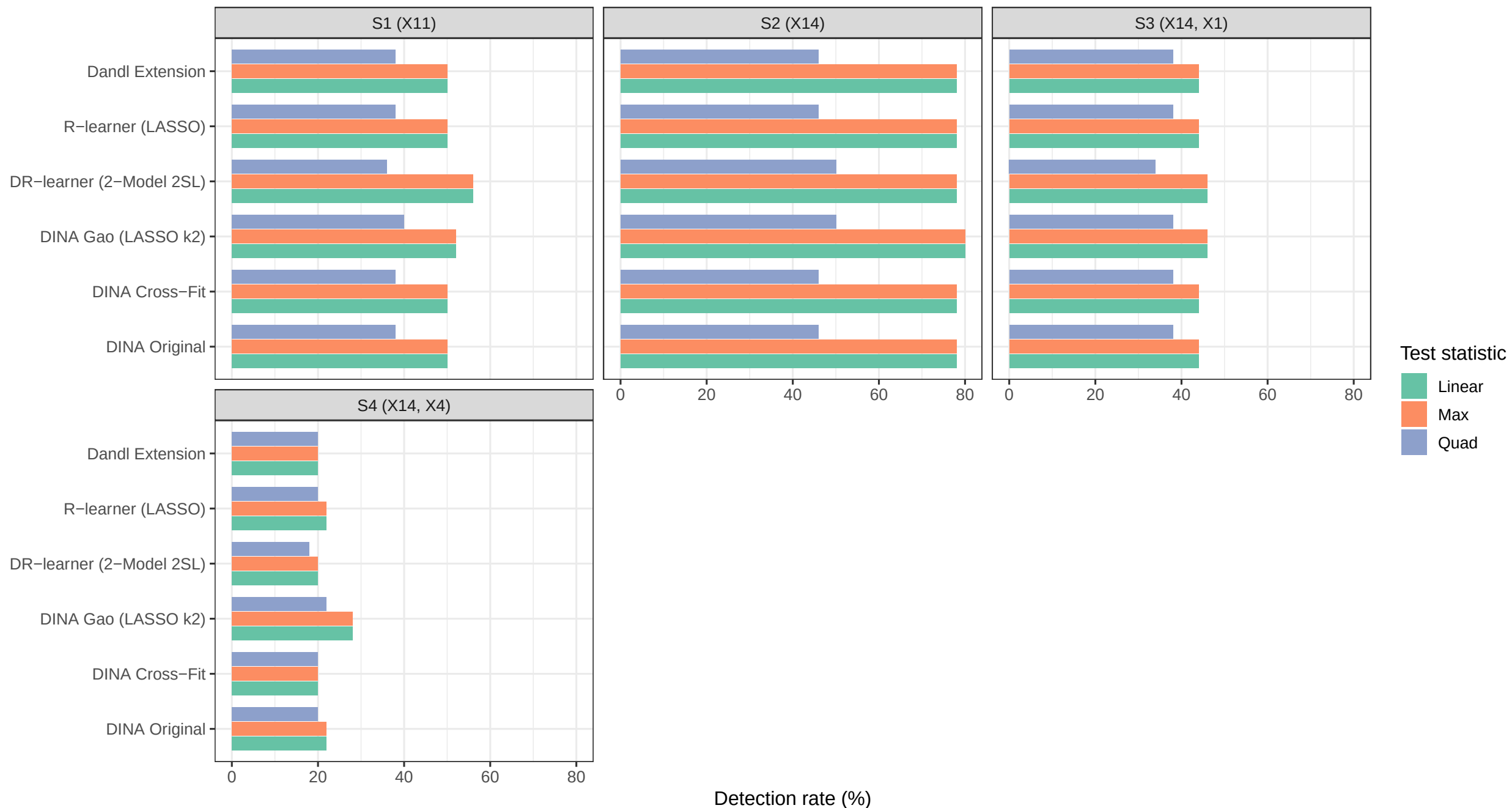


beta=0 = type-I error (target ~5%); beta>0 = power. Dashed line = 5%.



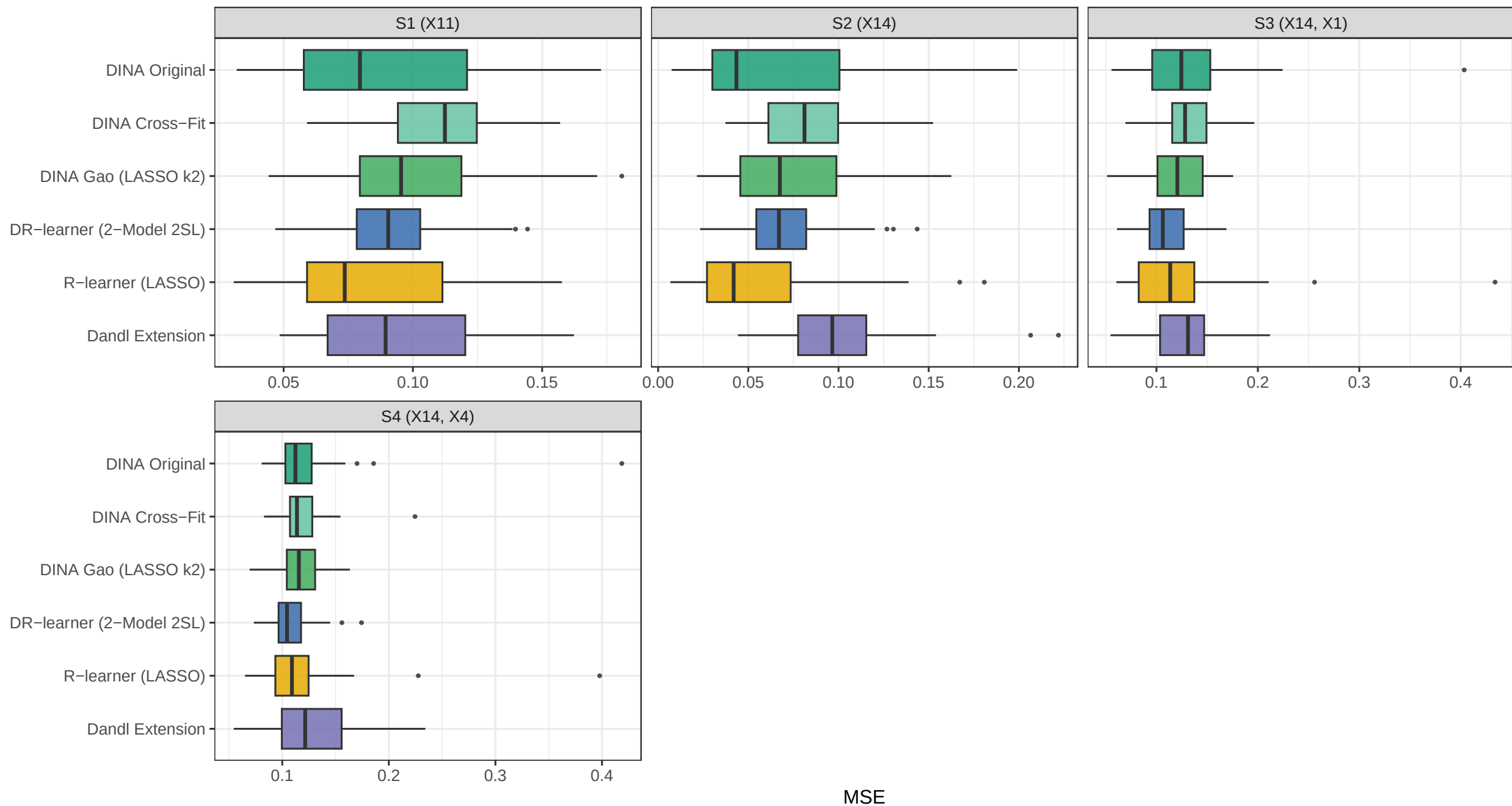
Detection rate by test statistic (beta = 2)

Linear vs Max vs Quadratic heterogeneity test, strongest signal.



Per-simulation MSE distribution (beta = 2)

Spread across the Monte-Carlo replications.



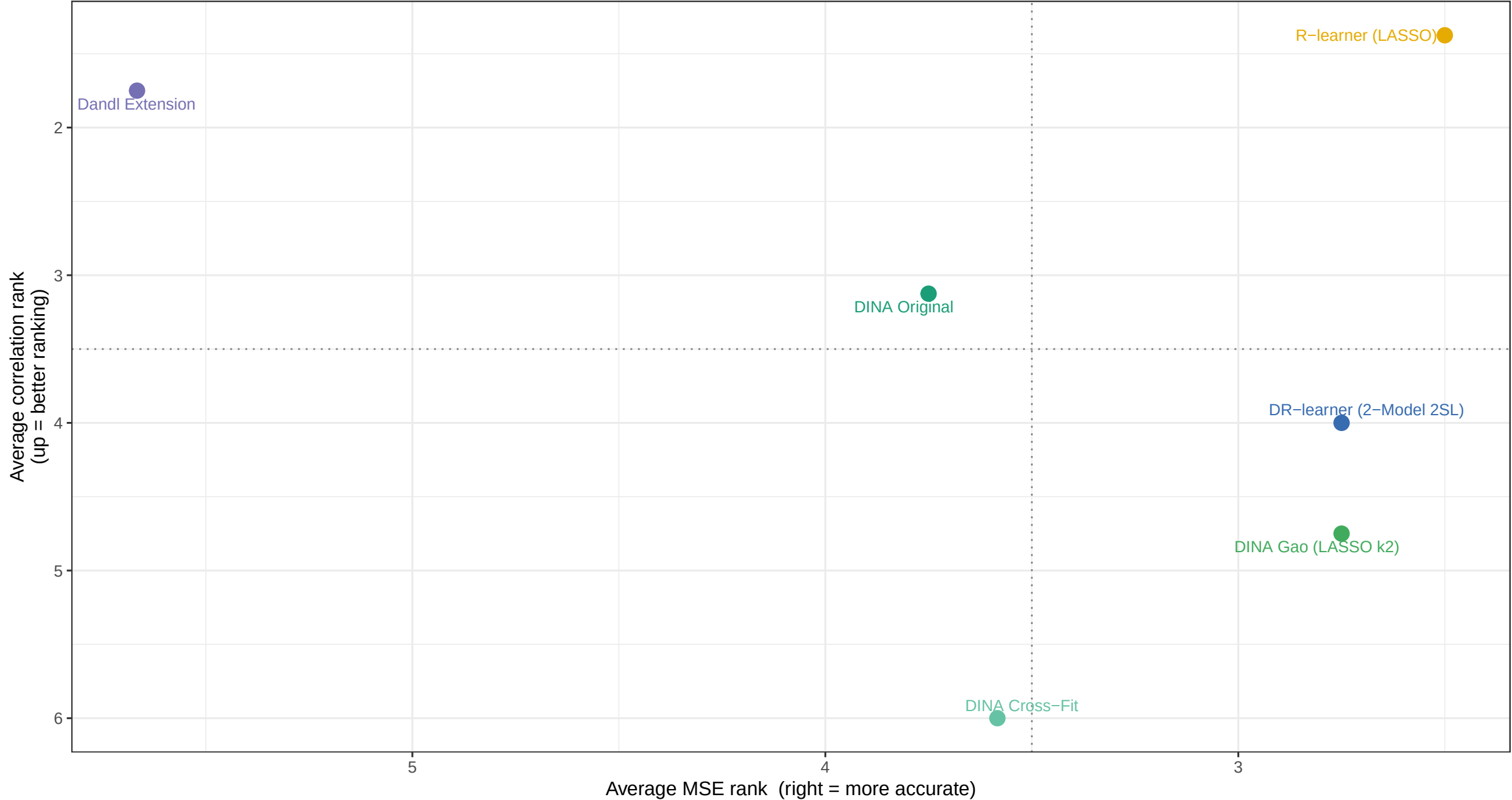
Cross-scenario MSE: rank within each scenario x beta cell

Green = best (lowest MSE), red = worst. Numbers are the mean MSE.



Accuracy vs ranking-quality tradeoff (avg rank across all cells)

Up & right = better on both. X: MSE magnitude (lower rank better). Y: correlation with true CATE (beta>0).



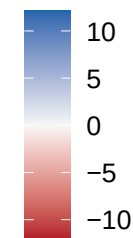
Pairwise head-to-head: net win margin (ROW minus COLUMN)

Avg over all scenarios & beta. Blue (+) = row beats column more often; most pairs tie in ~70–95% of sims.

Row method



Net win %
(row - col)

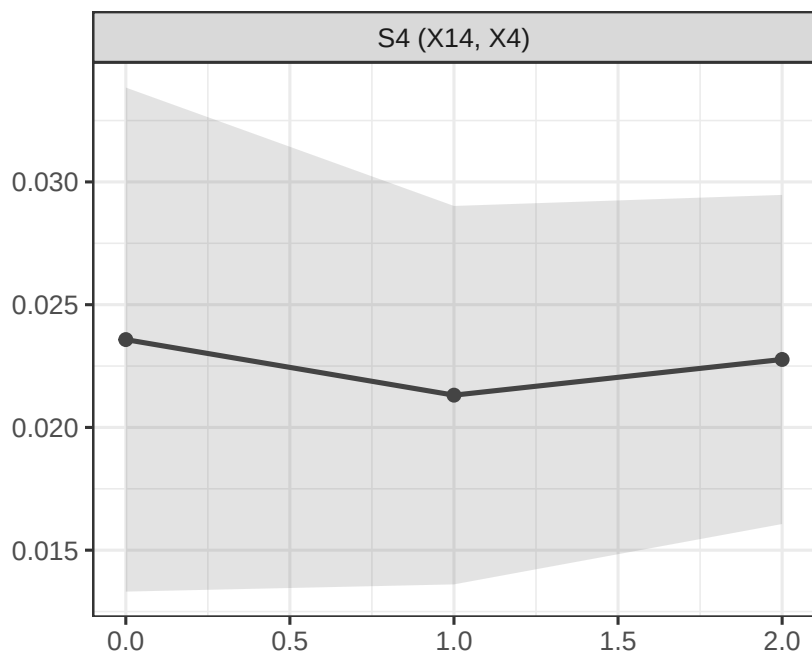
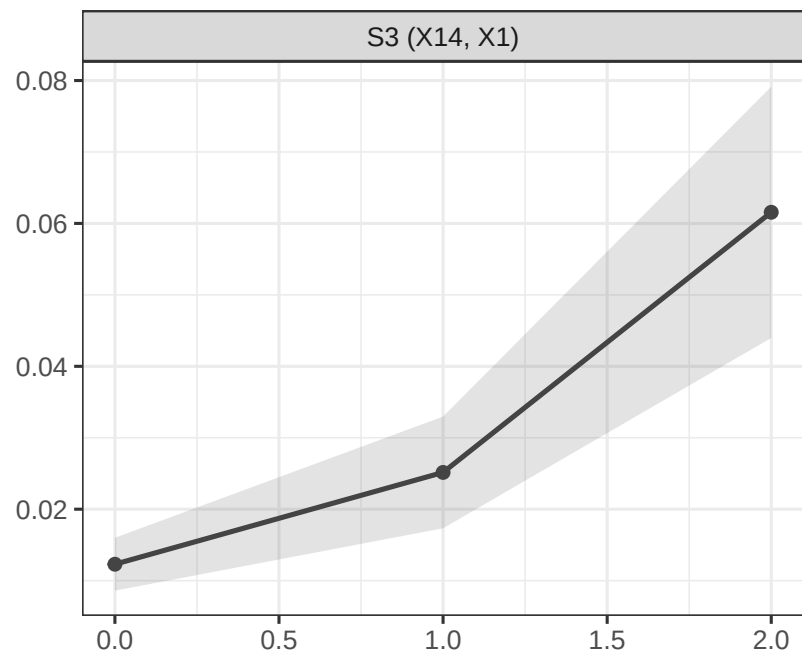
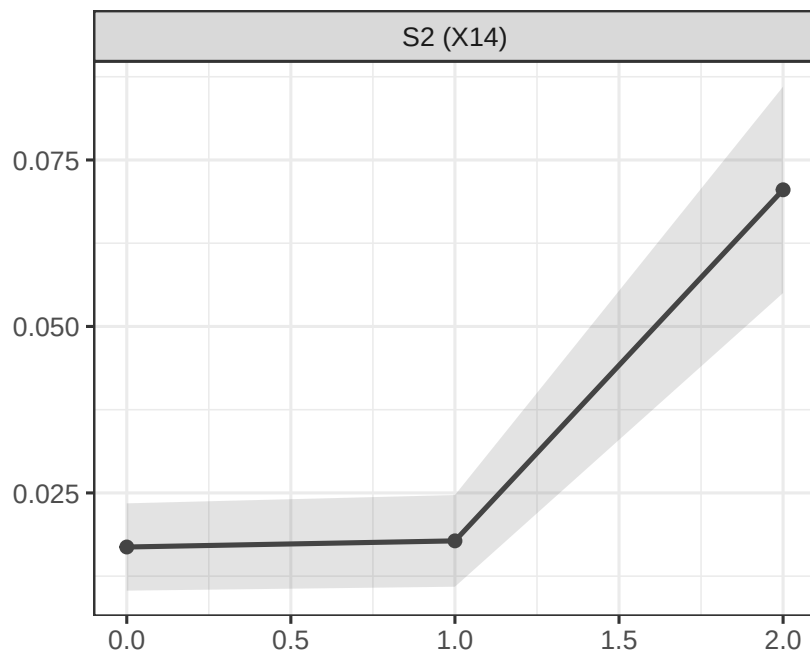
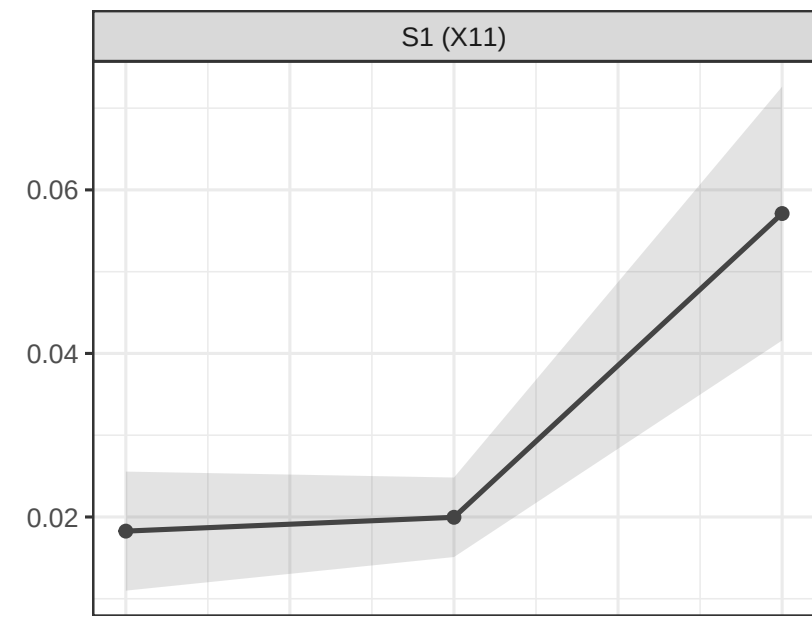


Column method

TMLE (Li 2026 VTE): estimated variance of treatment effect

VTE grows with beta where heterogeneity is real. Ribbon = 95% MC CI.

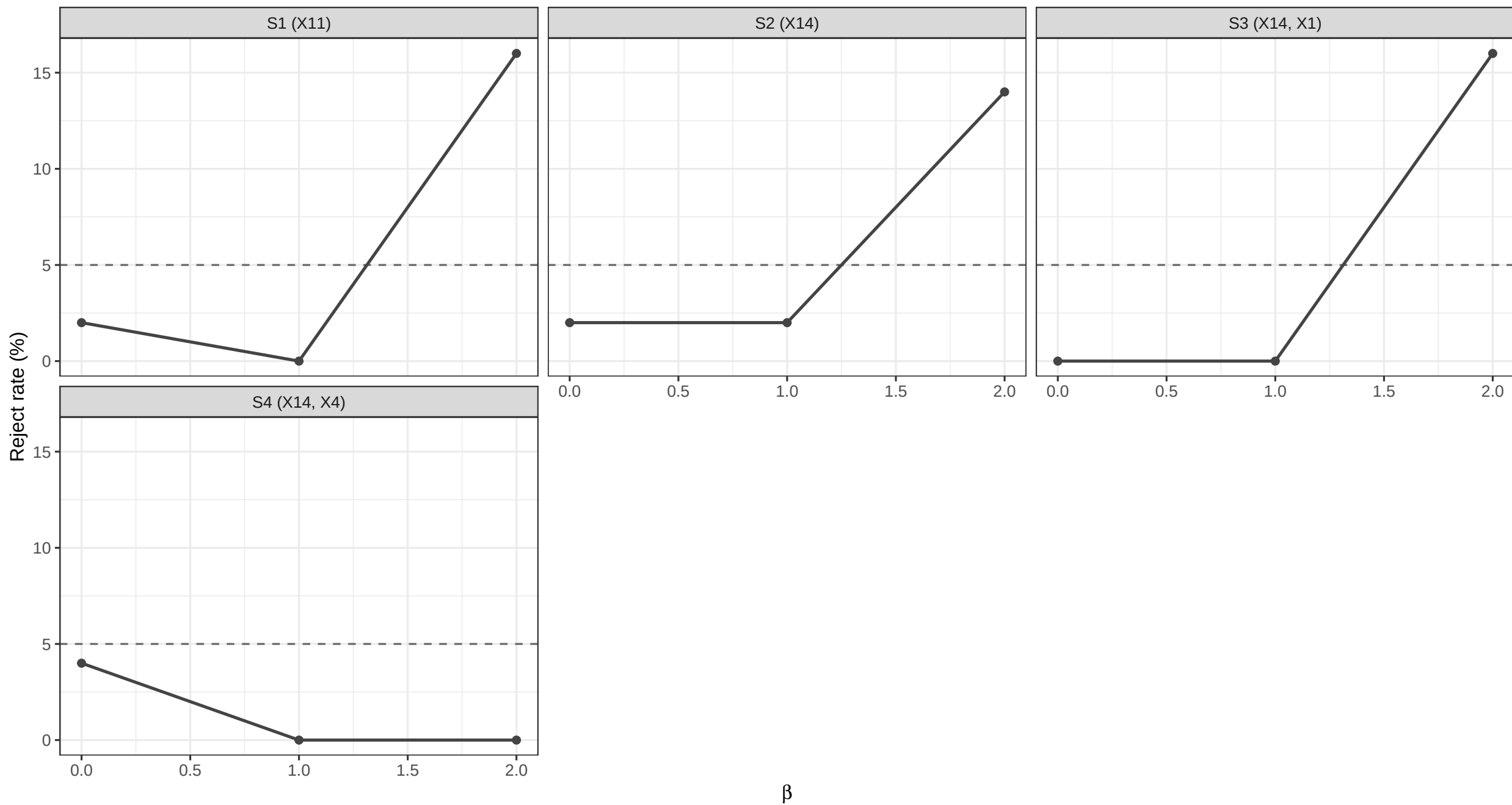
Mean VTE



β

TMLE (Li 2026 VTE): rejection rate (heterogeneity test)

beta=0 = type-I error (target ~5%, dashed); beta>0 = power.



Mean runtime per fit

